

Hip Complex

Introduction :-

- It is also known as coxofemoral jt.
- It articulating with acetabulum of the pelvis and head of the femur.
- It is a diarthrodial ball & socket jt.
- It has 3° freedom

Flexion/Extension - Sagittal plane

Abduction/Adduction - Frontal plane

Medial/Lateral rotatⁿ - Transverse plane

- Primary function of hip complex is to support the weight of the head, arms and trunk (HAT) Both in static and dynamic erect position.

Articular Surface :-

Proximal Articular Surface :-

- Cuplike concave socket of the hip jt is acetabulum. Located on the lateral aspect of the pelvic bone.

- 3 bones form pelvis

Ilium

Ischium

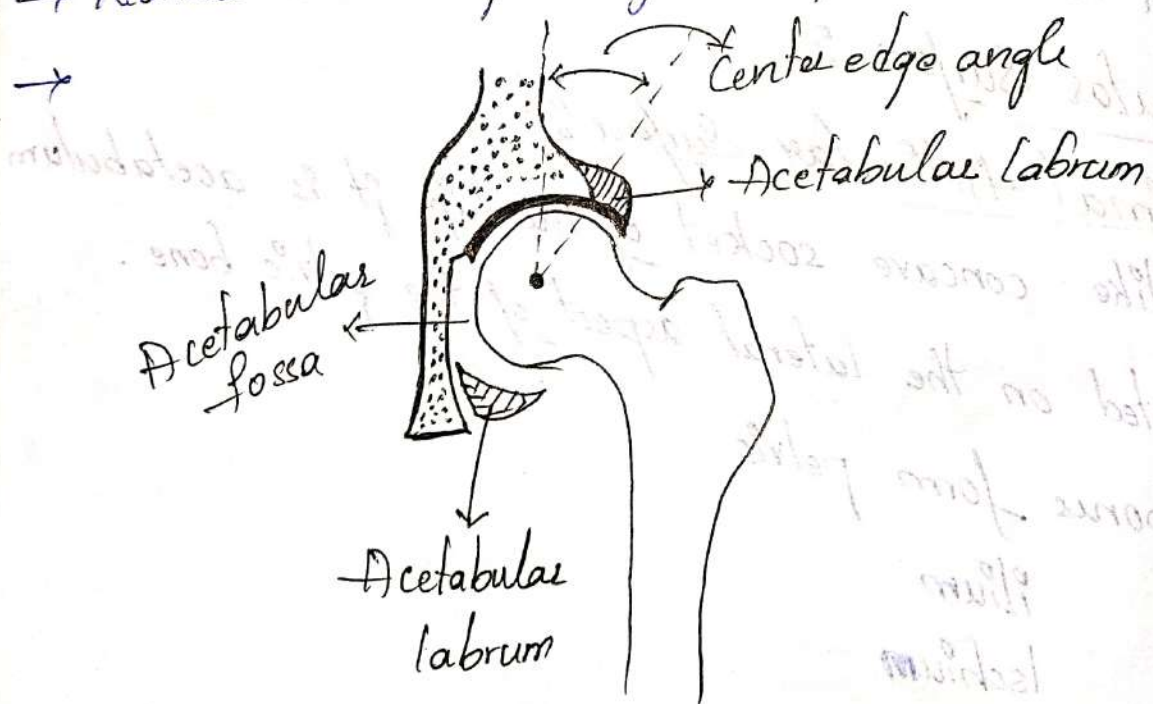
Pubis

- Pubis - $\frac{1}{5}^{\text{th}}$

Ischium - $\frac{2}{5}^{\text{th}}$

Ilium - Remainder

- Periphery of acetabulum (lunate surface) is covered with hyaline cartilage.
- Horseshoe-shaped area of cartilage articulates with the head of the femur.
- Inferior aspect of the lunate surface is interrupted by a deep notch called the acetabular notch.
- Acetabulum is laterally, inferior & anteriorly tilted.
- Acetabulum depth can be measured as the center edge angle of Wiberg.
- It is formed by the line connecting the lateral ring of the acetabulum and vertical line from the femoral head.
- Normal center edge angle in person is 25° .



- The entire periphery of the acetabulum is summed by a ring of wedge-shaped fibrocartilage called the acetabular labrum.

→ Acetabulum labrum increases concavity and stability.

Distal Articular Surface:-

→ Head of the femur rounded and covered with hyaline cartilage.

→ $\frac{2}{3}$ rd part of acetabulum is filled with femur.

→ Femoral head is small, roughened pit called fovea or fovea capitis.

→ Femoral head is medially, superiorly and anteriorly.

Angle of Inclination:-

→ It occurs in frontal plane b/w an axis through the femoral head & neck & longitudinal axis of the femoral shaft.

→ B/w 110° to 144° in adults

→ Angle of inclination ↓ in ~~females~~ females

Angle of inclination ↑ in females.

→ Coxa Valgus/Valga - ↑ the angle of inclination.

Coxa vara - ↓ the angle of inclination.

Angle of torsion:-

→ An axis through the femoral head and neck in the transverse plane will lie at an angle to an axis through the femoral condyle.

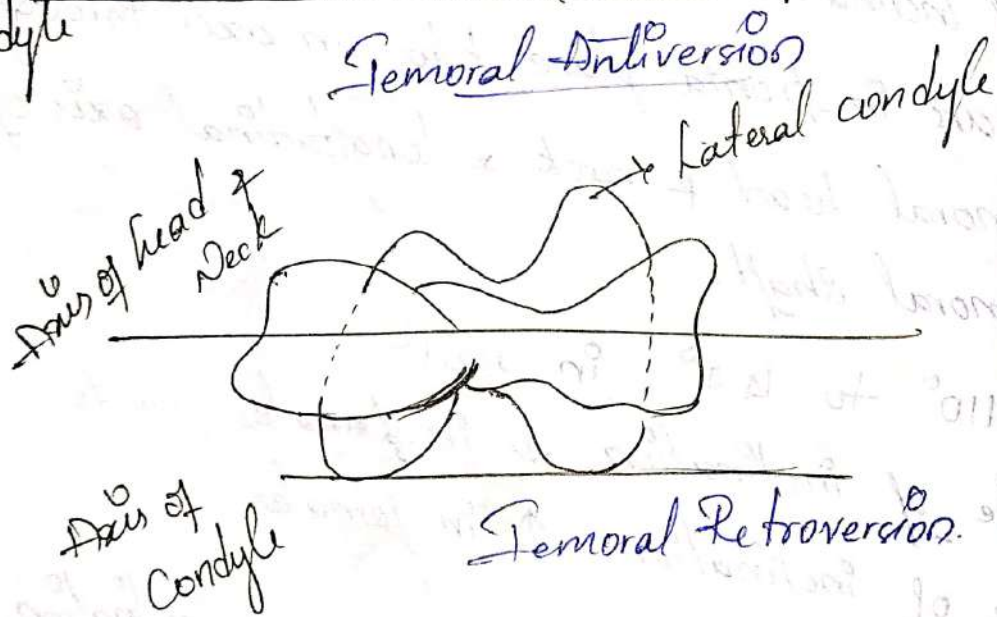
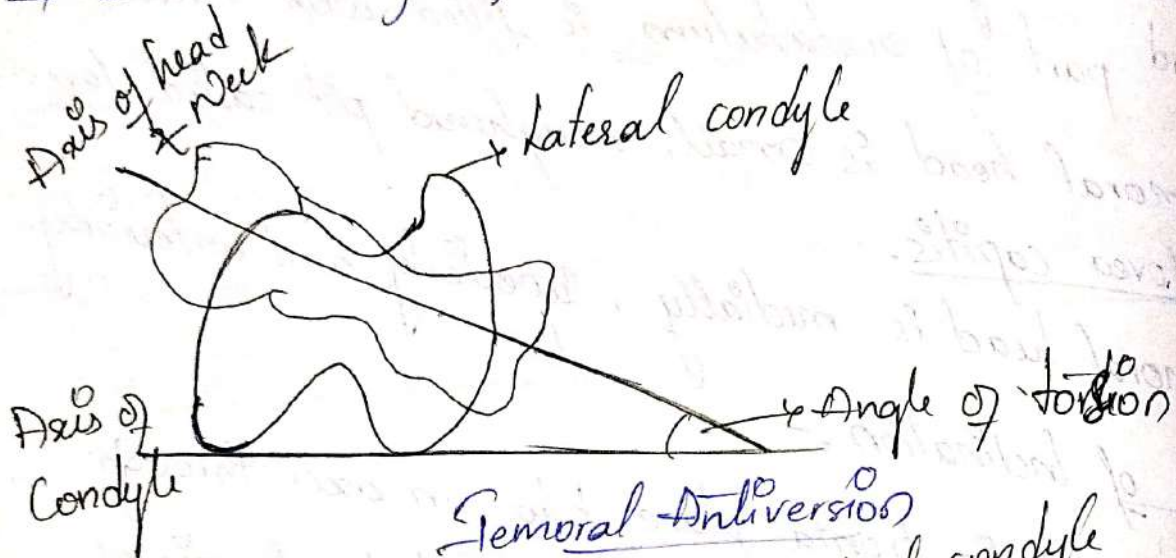
→ Normal angle of torsion - 10° - 20°

18° in females

15° in males.

→ Increased angle of torsion - Femoral Anteversion

→ Decreased angle of torsion - Femoral Retroversion



Articular Congruence :-

→ Articular contact b/w the femur and the acetabulum can be red in the non-weight bearing hip jt by a combinatⁿ of flexion, abduction & slight lateral rotatⁿ is known as Prog-Leg Position.

→ Acetabulum fossa may be imp in setting up a partial vacuum in the jt so that atmospheric pressure

contributes to stability by helping maintain contact b/w the femoral head & the acetabulum.

It capsule and Ligament :-

Joint Capsule -

- + It is an irregular, dense fibrous structure with longitudinal and oblique fibers.
- + Proximally - it articulates with acetabulum labrum
- + Fiber present below the femoral head is - Zona Orbicularis.
- + Distal - Femoral Neck.
- + Anteriorly - Thickened
- + Anteriorly - thin & loosely attached.
- + Proximally - Superior & inferior gluteal arteries supplied the hip capsule
- + Distally - Medial & lateral circumflex arteries supplied to the capsule.

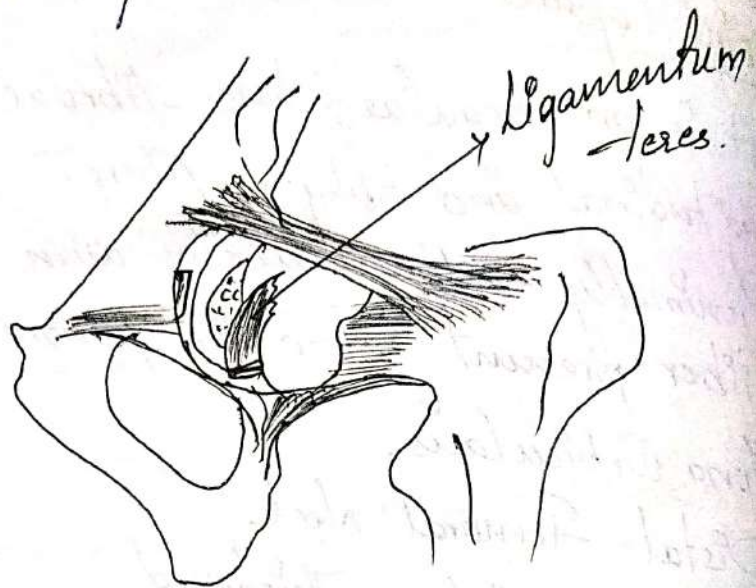
Ligaments -

- 1) Ligamentum teres
 - 2) Pubofemoral ligament
 - 3) Iliofemoral ligament
 - 4) Ischiofemoral ligament - Posterior Ligament
- } Anterior Ligaments

Ligamentum teres :-

- + It is an intra-articular but extrasynovial accessory structure.

- It is a triangular band that attaches to the acetabular notch.
- Bends with transvers acetabulum ligament & inserts on fovea capitis.



Ligamentum femoral:

- Fan-shaped ligament an inverted Y.
- Also known as Y Ligament of Bigelow.
- Apex of Y is attached to anterior-inferior iliac spine.
- 2 Arms are attached to along the intertrochanteric line of the femur.

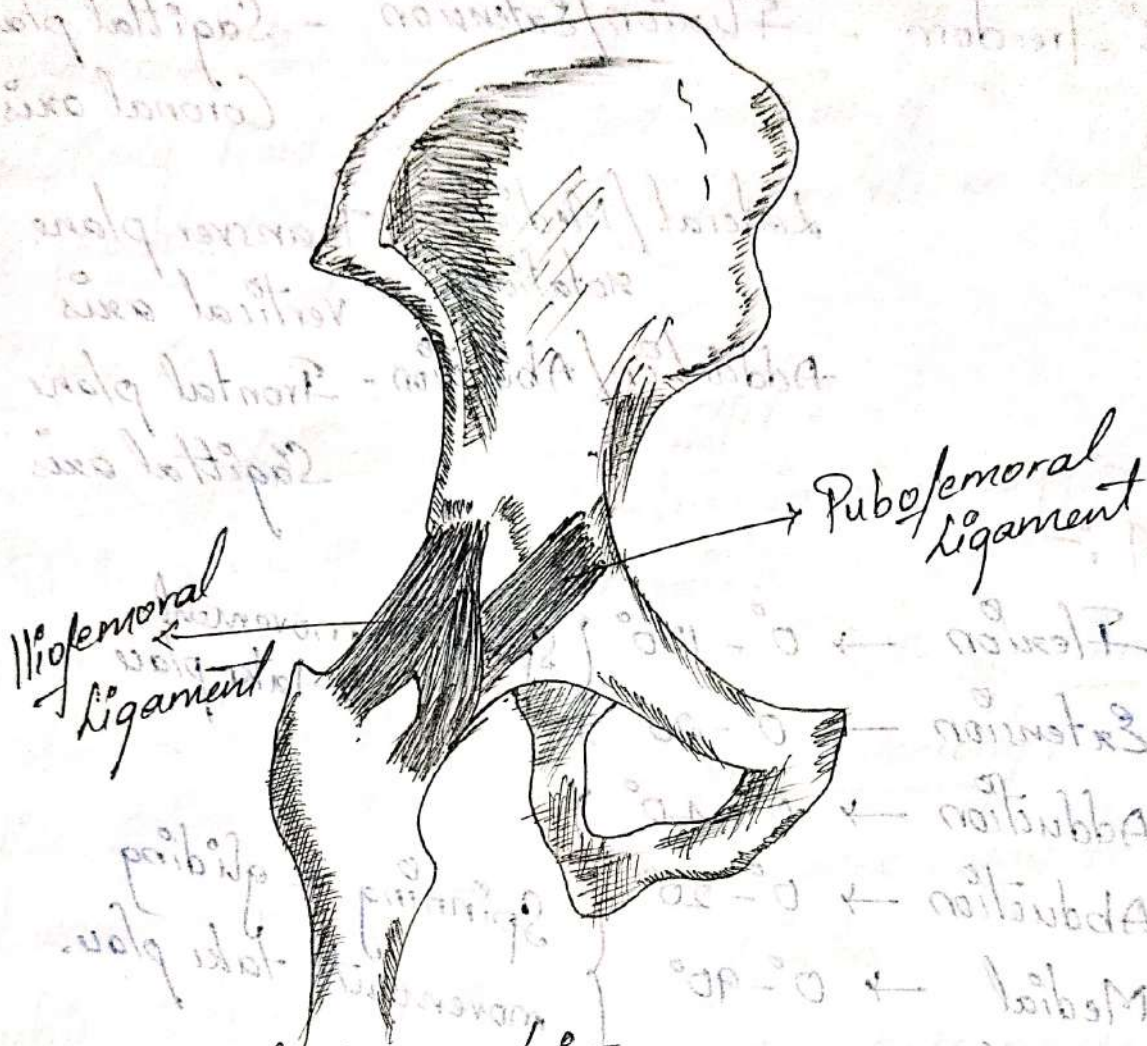
→ Superior band is the strongest and thickest of the hip ligaments.

Pubofemoral ligament:

- Arises from the anterior aspect of the pubic ramus & passes to the anterior surface of the

Intertrochanteric fossa.

→ Control lateral rotation in an extended position.



Ischiofemoral Ligament:-

→ Attaches to the posterior surface of the acetabular rim and the acetabulum labrum.



Function:

Osteokinematics -

→ 3° freedom - Flexion/Extension - Sagittal plane
Coronal axis

Lateral/Medial rotations - Transverse plane
vertical axis

Adduction/Abduction - Frontal plane
Sagittal axis

ROM :-

Flexion → $0^{\circ} - 120^{\circ}$
Extension → $0^{\circ} - 20^{\circ}$ } Spinning movement take place

Adduction → $0^{\circ} - 40^{\circ}$
Abduction → $0^{\circ} - 20^{\circ}$
Medial → $0^{\circ} - 90^{\circ}$
Lateral → $0^{\circ} - 90^{\circ}$ } Spinning & gliding movements take place.

→ The femoral head glide within the acetabulum in the direction opp to the motion of the distal end of the femur.

→ Flexion & Extension - Spinning movement occur.

→ Flexion - femoral head spins posteriorly

→ Extension - femoral head spins anteriorly.

→ Hip extension with complete knee flexion it is limited because of the muscle rectus femoris.

→ Add can be limited by graciles.

→ Add is limited by tensor fascia lata.

Motion of the pelvis on the femur :-

→ Whenever, the hip is wt. bearing, the femur is relatively fixed, and, in fact, motion of the hip is produced by movement of the pelvis on the femur.

Tilting :-

Anterior and Posterior pelvic tilt :-

→ It is a motion of the pelvis in the sagittal plane around a coronal axis.

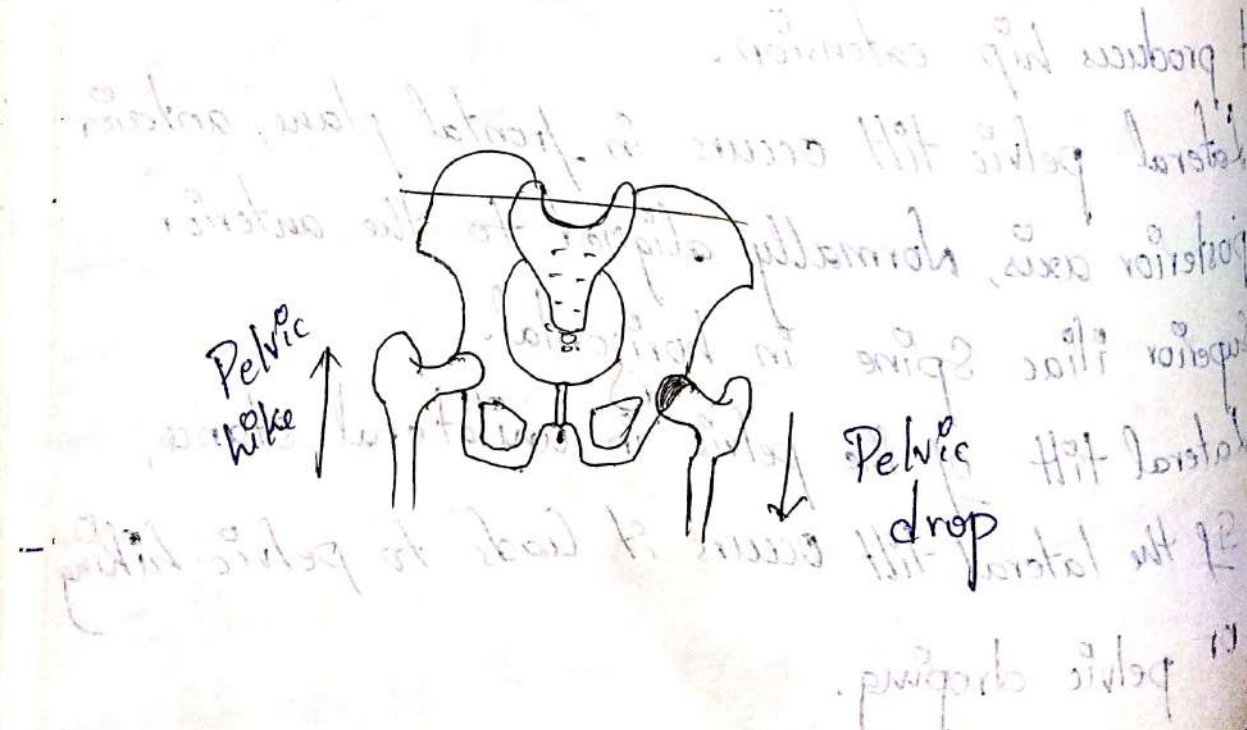
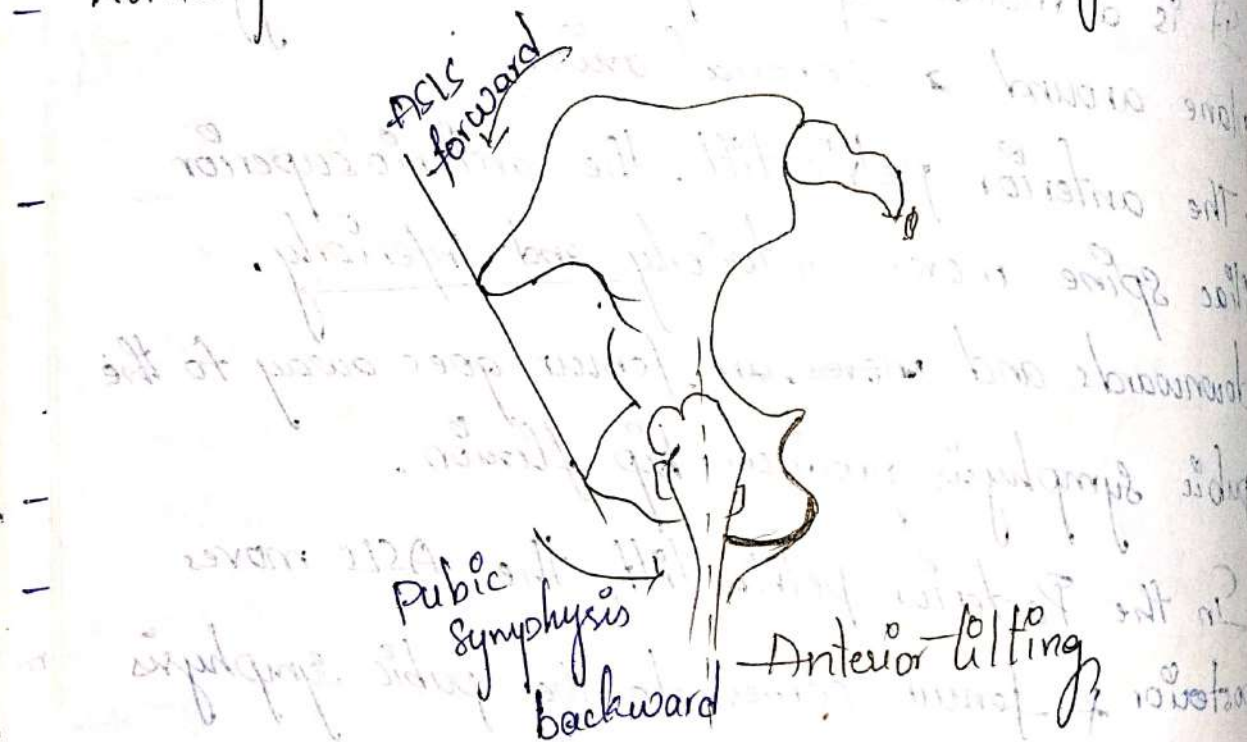
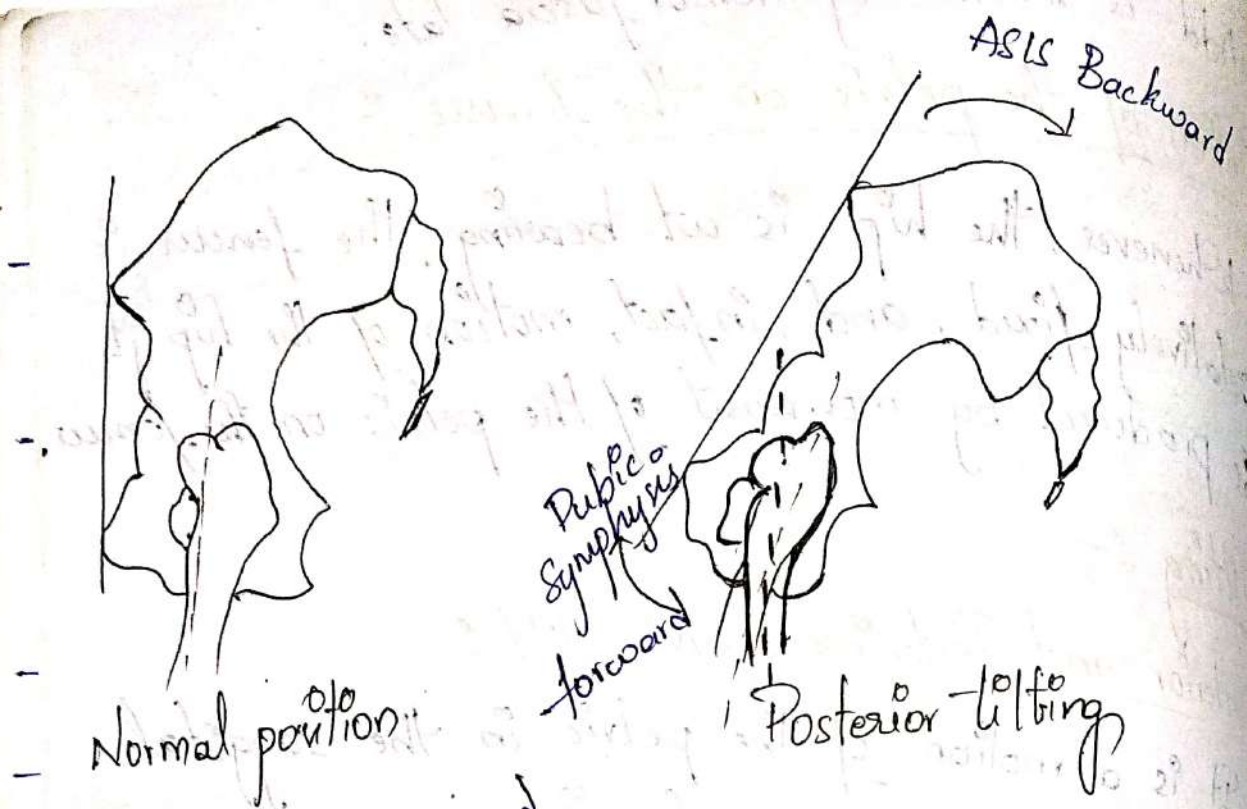
→ The anterior pelvic tilt, the anterosuperior iliac spine moves anteriorly and inferiorly, downwards and ~~more~~ as femur goes away to the pubic symphysis produces hip flexion.

→ In the Posterior pelvic tilt, the ASIS moves posterior & femur comes to the pubic symphysis it produces hip extension.

→ Lateral pelvic tilt occurs in frontal plane, anterior posterior axis, normally aligned to the anterior superior iliac spine in horizontal.

→ Lateral tilt of the pelvis in unilateral stance,

→ If the lateral tilt occurs it leads to pelvic tilting or pelvic dropping.



Lateral Shift of the pelvis:-

- Lateral pelvic tilt occur in bilateral stance.
- If both the limbs are wt. bearing, Lateral pelvic causes the pelvic shift to one side or the other.
- Pelvic shift, the pelvis cannot hike, it can only drop.
- If the pelvis will drop, the right hip it will be in adducted & the left hip it will be abducted.

Forward and Backward pelvic Rotation:-

- Pelvic rotation is the motion of the entire pelvic ring in the transverse plane, vertical axis.
- Forward or anterior rotation of the pelvis occur in unilateral stance when the side of the pelvis opp to the wt. bearing hip moves anteriorly.
- Forward rotation of the pelvis produce medial rotation of the wt. bearing hip joint.
- Backward rotation of the pelvis produces or occurs when the side of the pelvis opp to the wt. bearing hip moves posteriorly.
- Backward rotation of the pelvis produce lateral rotation of the supporting hip jt.

Arthokinematics

Open kinematic chain :-

→ Pelvic ^{femoral} forward chain motion :

When the femur, pelvis and spine moves in a coordinated way it is a open kinematic chain motion / pelvic femoral motion / pelvic femoral rhythm.

→ Here the ft serves as either end of the

Example : moving the head & arm ^{through} space.
moving of the foot ^{through} space.

Closed kinematic chain :-

→ The ft of the rt & lt lower limbs are part of a true closed chain, when the both lower limbs are wt-bearing & chain is defined as all the segments b/w the right foot, up through the pelvic & down through the left foot.

→ For hip to closed chain in standing, both ends of chain must be fixed. The head must often remains upright & vertically oriented during upright activities.

→ The movement at one ft create movement in atleast one another linkage in chain.

Muscles of Hip Jt :-

1^o Flexion - Flexors muscles

Iliopsoas → Origin - Iliac fossa
Insertion - Femur

Rectus femoris → O - ASIS
I - Tibial tuberosity

Tensor fascia lata → O - Anteriolateral lip
of iliac crest
I - Iliotibial band

Sartorius → O - ASIS
I - Anterior aspects of femur
to insert to the upper
portion of the medial
aspect of the tibia.

2^o flexion - Pectineus
Adductor longus
Adductor magnus
Gracilis

Adductors - Pectineus
Adductor longus
Adductor Magnus
Adductor Brevis
Gracilis.

Extensors - Gluteus maximus
Hamstrings

Abductors - Gluteus Medius [pelvis stabilizer]
(Rotator cuff of hip) Gluteus minimus [femoral head
stabilizers]

Lateral Rotators - Obturator Internus & Externus
Gemelli Superior & Inferior
Quadratus femoris
Piriformis

Medial Rotators -

There are no primary muscles

2^o muscles - Anterior fiber of Gluteus medius
Gluteus minimus
Tensor fascia lata.

Forces acting on hip Jt :-

→ Bilateral stance - $WR = WL$

Gravity movement = $WR \times DR = WL \times DL$

→ Unilateral stance - $HAT + HL$
 $\frac{2}{3}^{rd}$ $\frac{1}{6}^{th}$

Left leg Compression = $\left[\frac{2}{3}^{rd} \text{ of body wt} \times \frac{1}{3}^{rd} \text{ of HL (Rt)} \right]$
(HAT)

Hip Jt forces & Muscle function in stance :-

→ In frontal plane during bilateral stance the HAT wt is transmitted to sacroiliac jt & pelvis to the right & left femoral head.

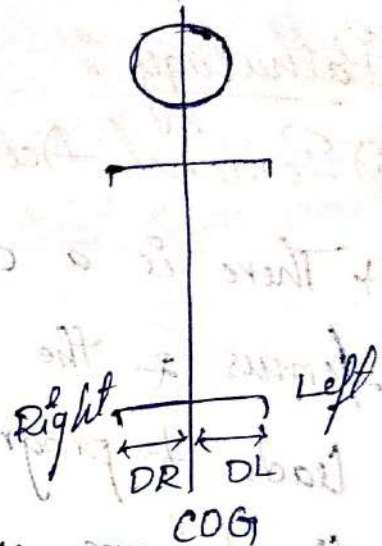
→ The jt axis of each hip lies on equal distance from the line of gravity of HAT.

∴ Gravitational movement arm of the rt hip is (DR) and left hip (DL)

$$WR = WL$$

→ The gravitational torque is $WR \times DR = WL \times DL$
Hence pelvis is maintained in equilibrium without assistance of acting muscle movement.

→ When bilateral stance is not symmetrical, in frontal plane, muscle activity will be necessary either to control from side to side motion or to return the hip to ~~upright~~ symmetrical stance.



→ The pelvis is shift to rt. Hip there is relative adduction of rt. hip & abduction of lt. hip.

→ To return to neutral position the active contraction of Rt. hip abductor would be expected

Unilateral Stance :-

→ If the left. leg is lifted & the HAT wt is being supported by Rt. hip it is in adduction with the lt. limb. ∴ Hanging body

→ The magnitude of wt. compressing the rt. hip is in rt. unilateral stance.

Rt. Leg Compression $\left[\frac{1}{6}^{th} \times W \right] + \left[\frac{2}{3}^{rd} \times W \right]$

HLW HAT

[Hanging Leg]
wt

Total compression of Rt. hip = $\frac{5}{6}^{th} \times W$

Pathology:

① Femoral-Acetabular Impingement (compress)
→ There is a disfunctional abutment of proximal femur & the acetabulum & results in pain & leads to progressive degeneration specifically at the labrum.

a. ~~Canning~~ ^{impingement} Impingement -

It occurs abnormal widening of femoral neck.

b. Pincer Impingement -

Overcovering of acetabulum on the femoral head

② Osteoarthritis

→ It is defined of the deterioration of articular cartilage and the changes in the articular tissue.

→ The presence of labral tear is an important factor in development of Osteoarthritis of hip.

③ Fracture - Avascular Necrosis